

Practical Data Analysis: Handling Null Values and Outliers

SECTION 1 — NULL VALUES

1.1 Definition

A **null or missing value** occurs when no data is recorded for an observation in a specific variable. Missing values occur for various reasons: a customer may skip a non-mandatory survey question, a system timeout might fail to log a transaction ID, or a field might not be applicable.

It is vital to distinguish between different types of absence:

- **Missing:** The data exists in reality but was not captured.
- **Zero:** A measured numerical value of 0.
- **Unknown:** The truth is completely unverifiable.
- **Not Applicable:** The field logically cannot have a value.

Sometimes, missingness itself contains information. Consider a dataset of 5,000 orders where `Discount_Amount` is missing for 400 orders. An analyst must investigate: How many are missing? Why? Is it meaningful? If a customer didn't use a code, the missingness signifies the lack of a discount, which can legitimately be mapped to ₹0.

1.2 When to Keep Null Values

An analyst should keep a value missing when replacing it would distort reality or create misleading metrics. Situations include:

- A newly launched product has no rating because nobody has rated it yet.
- A delivery date is unavailable because the order is still in progress.
- A field is genuinely not applicable to that specific record.

Worked Example: A logistics table records `Order_Date` and `Delivery_Date`. For 150 active orders, `Delivery_Date` is missing. If an analyst replaces these nulls with today's date or an arbitrary zero, the calculation for "Average Delivery Days" will be instantly corrupted. The values must remain null because the event (delivery) has not yet occurred.

1.3 When to Remove Null Values

Removing observations containing missing data should not be an automatic reflex. An analyst must carefully evaluate:

- The amount of missing data relative to the total dataset size.
- The importance of the variable to the final objective.
- Whether the record can still be used for other valid analyses.
- Whether removing the observations introduces bias into the dataset.

Before/After Example: Out of 100,000 completed orders, 12 rows are missing the Payment_Method. If the goal is strictly to analyze payment method preferences, these 12 rows cannot be evaluated. Because 12 observations constitute a negligible 0.01% of the dataset and their removal will not introduce meaningful bias, the analyst safely removes these specific records before calculating the distribution.

Crucial rule: Do not remove observations simply because they contain a null value.

1.4 When to Impute Null Values

Imputation is the process of replacing a missing value with a statistically or contextually reasonable estimate. Like removal, imputation is a decision that requires contextual judgment.

Mean Imputation

The mean (average) is calculated by summing all available values and dividing by the count of those values. Mean imputation is appropriate when the numerical data is relatively symmetric. However, it is highly sensitive to extreme values, which can distort the estimate, and its use artificially reduces the overall variability of the data.

- **Example:** Five orders have delivery times of 2, 3, 4, 3, and 3 days. One order is missing its delivery time. The mean is $15 \div 5 = 3$ days. Imputing 3 days is mathematically reasonable here.

Median Imputation

The median is the absolute middle value when the data is sorted. It is superior to the mean when the dataset is skewed or contains extreme observations.

- **Example:** Customer spending for five accounts in a segment are: ₹500, ₹600, ₹700, ₹800, and one massive purchase of ₹15,000.
Mean = $\text{₹}17,600 \div 5 = \text{₹}3,520$.
Median = $\text{₹}700$.

Using ₹3,520 to impute a missing customer's spend would falsely imply a high typical transaction. The median (₹700) accurately represents the typical customer.

Mode Imputation

The mode is the most frequently occurring category. It is primarily used for categorical variables.

- **Example:** A dataset shows Shipping Methods: Standard, Standard, Express, Standard, Next-Day. The mode is "Standard". If one record is missing shipping data, imputing "Standard" is a statistically reasonable guess, assuming missingness is random.

Method	Type of Variable	When Useful
Mean	Numerical	Relatively symmetric data
Median	Numerical	Skewed data / extreme values
Mode	Categorical	Most frequent category

These are general guidelines. Imputing data injects artificial certainty into a dataset, so analysts should thoroughly document their reasoning.

SECTION 2 — OUTLIERS

2.1 Definition

An outlier is an observation that is unusually distant from the general pattern of the data. Outliers can appear in order values, product prices, delivery days, or customer spending metrics.

Analysts must clearly distinguish between the possible causes of an outlier:

- **Genuine extreme observation:** A bulk buyer legitimately placing a massive order.
- **Data-entry error:** A user typing their age as 299 instead of 29.
- **Measurement/system error:** A corrupted sensor logging an impossible metric.
- **Rare business event:** A temporary flash sale causing an anomalous spike.

Strongly emphasize: An outlier is not automatically an error.

2.2 IQR Method (Position & Interpolation)

The Interquartile Range (IQR) method is a robust technique for detecting potential outliers. To calculate the quartiles accurately, analysts use the position formula $P_k = k(n+1)/4$, and apply linear interpolation when the position falls between two integers.

- **Q1 (25th percentile):** Position $P_1 = 1(n+1)/4$
- **Q2 (Median/50th percentile):** Position $P_2 = 2(n+1)/4$
- **Q3 (75th percentile):** Position $P_3 = 3(n+1)/4$
- **IQR:** $Q3 - Q1$
- **Lower Fence:** $Q1 - (1.5 \times IQR)$
- **Upper Fence:** $Q3 + (1.5 \times IQR)$

Complete Calculation Example (Outliers on Both Sides):

Dataset of $n=10$ order values (in ₹), sorted from lowest to highest:

15, 300, 320, 340, 360, 380, 400, 420, 450, 1500.

Step 1: Calculate Q1

Position $P_1 = 1(10+1)/4 = 11/4 = 2.75$

Since 2.75 is between the 2nd (₹300) and 3rd (₹320) values, interpolate by taking 75% of the distance between them:

$$Q1 = 300 + 0.75 \times (320 - 300) = 300 + 15 = ₹315$$

Step 2: Calculate Q2 (Median)

Position $P_2 = 2(10+1)/4 = 22/4 = 5.5$

Interpolate between the 5th (₹360) and 6th (₹380) values:

$$Q2 = 360 + 0.5 \times (380 - 360) = 360 + 10 = ₹370$$

Step 3: Calculate Q3

Position $P_3 = 3(10+1)/4 = 33/4 = 8.25$

Interpolate between the 8th (₹420) and 9th (₹450) values:

$$Q3 = 420 + 0.25 \times (450 - 420) = 420 + 7.5 = ₹427.5$$

Step 4: Calculate IQR and Fences

$$IQR = 427.5 - 315 = 112.5$$

$$\text{Lower Fence: } 315 - (1.5 \times 112.5) = 315 - 168.75 = ₹146.25$$

$$\text{Upper Fence: } 427.5 + (1.5 \times 112.5) = 427.5 + 168.75 = ₹596.25$$

Step 5: Identify Potential Outliers

Any value $< ₹146.25$ or $> ₹596.25$ is flagged.

- ₹15 is less than ₹146.25 (Lower Outlier).

- ₹1500 is greater than ₹596.25 (Upper Outlier).

2.3 When to Keep an Outlier

In many analytical contexts, the safest and most objective choice is to keep the outlier, especially when its origin cannot be definitively classified as an error. If an observation represents a factual, albeit rare, event, it belongs in the dataset.

For instance, an unusually massive ₹75,000 wholesale purchase might be flagged mathematically. Removing it without business alignment could violently distort total revenue reporting and ignore highly valuable customer behaviors. **Analysts should consult with domain experts (like sales managers) before deciding to exclude legitimate extremes.**

2.4 When to Remove an Outlier

Deciding to permanently remove an observation requires solid justification and consensus. It is typically considered only when an observation is definitively proven to be a measurement error, an impossible value, or a corrupted record that cannot be salvaged.

- **Original Data:** Package weight recorded as 450 kg for a smartphone case.
- **Detection:** Value far exceeds the upper fence.
- **Investigation:** A smartphone case physically cannot weigh 450 kilograms.
- **Action:** If the true weight (e.g., 0.45 kg) cannot be confirmed with operations, the analyst may consult stakeholders to safely remove the record from weight-based logistics modelling.

Do NOT remove an observation merely because it lies outside the IQR fences. Fences are highlighters, not decision-makers.

2.5 When to Impute an Outlier Using Median

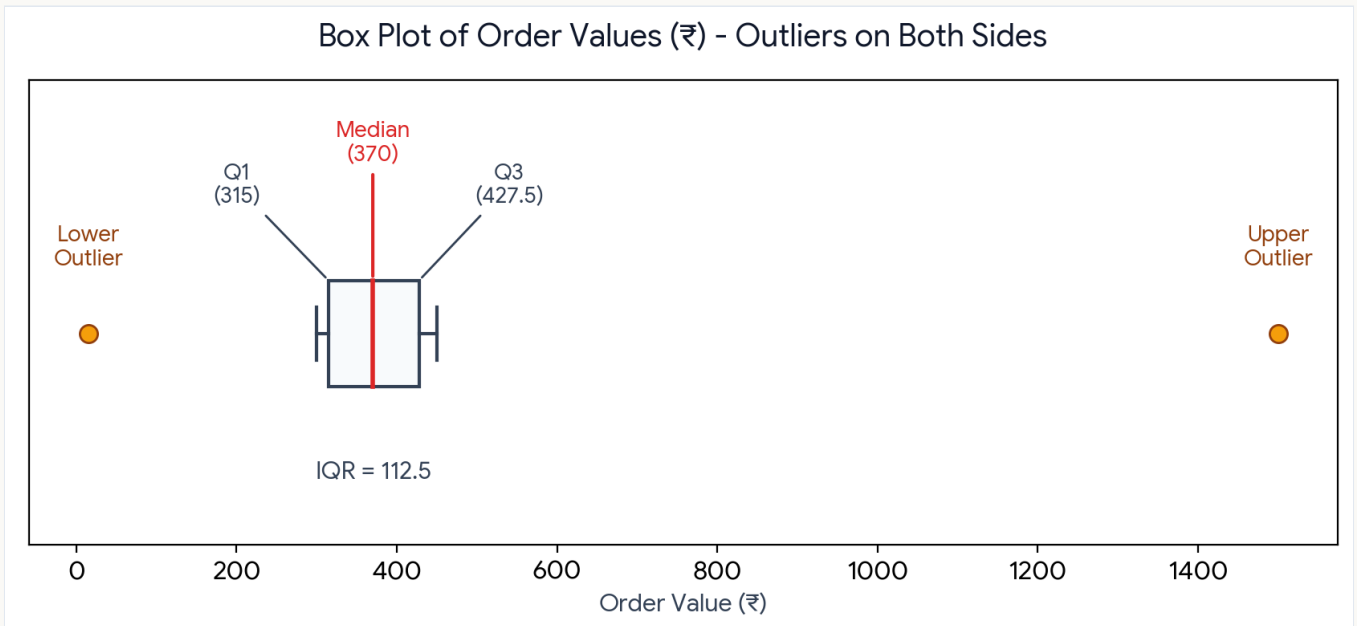
Imputing an outlier is a sensitive intervention because it modifies the original data footprint. It should be applied carefully, often as a "last resort" when preserving the rest of the row's data is critical, but the specific metric is known to be flawed.

Consultative Example:

- **Original observation:** Customer spent ₹9,999,999 on a standard order.
- **Verification:** The IT team confirms a known system glitch duplicated the '9' key. The true value is lost.
- **Decision:** The analyst confers with the data team. Removing the whole row loses valuable demographic data. They agree to impute.
- **Replacement:** The median spend for similar orders (₹1,200) is used instead of the mean, ensuring the glitch itself doesn't inflate the replacement value.

2.6 Box Plot

A box plot is a visual representation of the IQR method, allowing analysts to quickly grasp the centre, spread, skewness, and extremes.



- **Q1 & Q3:** The left and right edges of the central box.
- **Median:** The prominent vertical line inside the box.
- **IQR:** The entire width of the box.
- **Lower Whisker:** Extends to the lowest data point that is $\geq$ the lower fence (in our example, ₹300).
- **Upper Whisker:** Extends to the highest data point that is $\leq$ the upper fence (in our example, ₹450).
- **Potential Outliers:** The isolated points plotted beyond the whiskers (₹15 and ₹1500).

FINAL — ANALYST DECISION FRAMEWORK

Null Values

Identify → Measure → Understand why → Consult Domain Knowledge → Keep / Remove / Impute

Outliers

Detect → Investigate → Seek Consensus → Keep / Correct / Remove / Impute

FINAL CHEAT SHEET

Situation	Analyst Thinking	Collaborative Action
Meaningful missing value	Missingness represents business reality	Keep (Document context)
Non-critical missing data	Negligible bias risk	Remove (With transparency)
Numerical, roughly symmetric	Central tendency is stable	Impute via Mean (If necessary)
Numerical, skewed/extreme	Mean is untrustworthy here	Impute via Median (If necessary)
Genuine extreme observation	Mathematically unusual, practically valid	Keep (Do not alter reality)
Confirmed system error	Data is demonstrably false	Correct or Remove (Consult IT/Ops)
Flawed data, row is needed	Need to salvage demographic info	Impute via Median (Document change)
IQR-flagged observation	Statistical anomaly found	Investigate (No automatic action)

- **Null does not automatically mean remove.**
- **Outlier does not automatically mean remove.**
- **IQR identifies potential outliers; it does not decide their treatment.**
- **When in doubt, it is safer to preserve the data as-is, consult stakeholders, and let business logic drive the final decision.**